Technical Architecture for Defensible Multi-Modal AI Detection in Academic Submissions

Forensic Text and Document Analysis

Establishing the provenance of student writing requires isolating the fundamental divergence between human cognitive composition and the statistical token sampling of large language models (LLMs). Generative models produce sequences by autoregressively sampling from learned conditional probability distributions, systematically optimizing for contextual coherence and sequence likelihood. Human composition, by contrast, is characterized by non-uniform information density, non-linear revision paths, idiosyncratic lexical variety, and mechanical document editing trails. Disciplinary defensibility requires an evidence chain that corroborates statistical text analysis with binary container forensics and deterministic document metadata.

Lightweight Statistical and Information-Theoretic Methods

Statistical text detection operates on the probability distribution of token sequences under a reference language model. To achieve zero per-submission operating expenditure, these algorithms must execute directly in the client browser using WebAssembly (WASM) or across serverless edge architectures constrained by strict execution time and memory limits.

Token Perplexity and Sequence Cross-Entropy

Token perplexity measures the degree of unexpectedness in a text sequence relative to an underlying reference probability model. For a tokenized text sequence $X = (x_1, x_2, \ldots, x_N)$, the negative log-likelihood (NLL) and corresponding perplexity (PPL) under model parameters θ are formalized as:

$$\mathrm{NLL}(X) = -\frac{1}{N}\sum_{i=1}^{N} \log P_\theta\left(x_i \mid x_1, x_2, \ldots, x_{i-1}\right)$$

$$\mathrm{PPL}(X) = \exp\big(\mathrm{NLL}(X)\big) = \prod_{i=1}^{N} P_\theta\left(x_i \mid x_{<i}\right)^{-\frac{1}{N}}$$

Machine-generated text concentrates along paths of low cross-entropy because nucleus, top-k, or temperature-scaled sampling algorithms disproportionately select high-probability tokens. Human writing displays substantial variance in sequence entropy, moving between predictable common transitions and semantically dense, domain-specific lexical clusters.

Sentence Burstiness and Informational Dispersion

Perplexity alone yields unacceptable false-positive rates when applied to uniform or formulaic human writing. Sentence burstiness resolves this limitation by evaluating the variance and dispersion of sentence-level perplexity across a document. Letting s_j represent the j-th sentence in a document containing Msentences, with $p_j = \text{PPL}(s_j)$, the sample mean μ_p and sample standard deviation $\sigma_p$are computed as:

$$\mu_p = \frac{1}{M}\sum_{j=1}^{M} p_j\,, \quad \sigma_p = \sqrt{\frac{1}{M-1}\sum_{j=1}^{M}(p_j - \mu_p)^2}$$

The bounded burstiness metric B maps this dispersion onto the interval $[-1,1]$:

$$B = \frac{\sigma_p - \mu_p}{\sigma_p + \mu_p}$$

Machine-generated passages exhibit low burstiness ($B \to -1$), displaying a flat, uniform cadence where sentence perplexity remains consistent across paragraphs. Human composition exhibits positive burstiness ($B > 0$), characterized by short, syntactically simple statements alternating with compound, semantically demanding constructions that produce sharp perplexity spikes.

Bin-Rank Distributions and Selection Entropy

The Giant Language Model Test Room (GLTR) methodology categorizes tokens based on their predictive rank under a baseline language model. Given a preceding context $x_{<i}$, the model calculates conditional probabilities across its entire vocabulary V, ordering tokens by probability rank $r(x_i) \in \{1,2,\ldots,|V|\}$. Tokens are then assigned to four discrete rank partitions:

- T_{top10}: $r(x_i) \leq 10$ (dominant green tier)
- T_{top100}: $10 < r(x_i) \leq 100$ (secondary yellow tier)

- T_{top1000}: $100 < r(x_i) \leq 1000$ (low-probability red tier)
- T_{tail}: $r(x_i) > 1000$ (outlier purple tier)

In machine-generated text produced via standard decoding parameters, the proportion of tokens falling within T_{top10} consistently exceeds 75% to 85%, while tokens situated in T_{tail} remain negligible. Conversely, human writing produces a heavy-tailed distribution where T_{top10} typically drops to $50\% - 60\%$, and T_{tail} reaches $8\% - 15\%$ due to topical phrasing, idioms, and irregular grammar.

Normalized Perplexity and Conditional Probability Curvature

Standard perplexity varies across diverse essay prompts and academic domains. The Binoculars architecture eliminates prompt sensitivity by calculating the normalized ratio between the perplexity of a primary observer model and the cross-perplexity derived from a secondary performer model:

$$\text{Score}_{\text{Bino}}(X) = \frac{\log P_{\mathcal{M}_1}(X)}{\log P_{\mathcal{M}_2}(X)} \approx \frac{\text{PPL}_{\mathcal{M}_1}(X)}{\text{CrossPPL}(\mathcal{M}_1, \mathcal{M}_2, X)}$$

When implemented with small open-source architectures (such as paired 1-billion-parameter Llama models), this cross-perplexity ratio isolates artificial regularity from prompt-specific vocabulary constraints.

Fast-DetectGPT strengthens this statistical foundation by measuring conditional probability curvature directly, avoiding the need for multi-step sampling perturbations. The framework evaluates the difference between the observed log-probability of a passage and its expectation over local conditional perturbations:

$$\tilde{d}(x, p_\theta, q_\phi) = \frac{\log p_\theta(x) - \mathbb{E}_{\tilde{x} \sim q_\phi(\cdot|x)}[\log p_\theta(\tilde{x})]}{\tilde{\sigma}}$$

Because machine-generated text samples occupy negative curvature regions (representing local probability peaks in the model's manifold), the metric $\tilde{d}$ yields pronounced positive values for synthetic passages while remaining centered near zero for human prose.

Lexical Diversity and Stylometric Metrics

Information-theoretic assessments are supplemented by non-neural stylometric algorithms to quantify lexical diversity:

- **Moving-Average Type-Token Ratio (MATTR):** To mitigate sample-length dependence, the system calculates vocabulary richness over sliding 50-token windows rather than standard whole-document Type-Token Ratio (V/N) or Root-TTR ($V/\sqrt{N}$).
- **Hapax Legomena Ratio (H_L):** Computed as V_1/N (where V_1 represents words appearing exactly once), this ratio identifies human academic writing by its elevated proportion of unique, single-use topical terms, in contrast to the repetitive closed-class connectors typical of synthetic text.
- **Simpson's Diversity Index (D_S):** Evaluated as $1 - \frac{\sum n_i(n_i-1)}{N(N-1)}$, this metric captures the probability that two randomly selected words represent distinct types, identifying synthetic homogeneity.

Metric	Human Academic Writing Baseline	Synthetic Text (LLM) Range	ESL / Non-Native Student Range	Primary Vulnerability / Failure Mode
Token Perplexity (PPL)	High (> 45.0)	Low to Moderate ($8.0 - 22.0$)	Moderate to Low ($14.0 - 28.0$)	Formal citations, technical definitions, and boilerplate prompts
Sentence Burstiness (B)	Positive (0.15 to 0.65)	Strongly Negative (-0.60 to -0.10)	Neutral (-0.05 to 0.25)	Highly structured procedural lab manuals
GLTR T_{top10} Proportion	$0.48 - 0.62$	$0.78 - 0.94$	$0.68 - 0.82$	Strict disciplinary jargon and formulaic prompts
Moving-Average TTR	> 0.72	$0.45 - 0.61$	$0.52 - 0.64$	Short submissions ($N <$ 200 tokens)

Metric	Human Academic Writing Baseline	Synthetic Text (LLM) Range	ESL / Non-Native Student Range	Primary Vulnerability / Failure Mode
Binoculars Ratio	Greater than decision threshold	Less than decision threshold	Borderline threshold values	Highly constrained analytical responses
Fast-DetectGPT ($\tilde{d}$)	≤ 0.15	≥ 1.20	$0.10 - 0.45$	Aggressive paraphrase transformations

Span-Level Neural Classifiers

Accurately evaluating hybridized submissions—where students merge AI-generated paragraphs into human-authored drafts—requires token-level and sentence-level neural classification.

Browser and Edge Classifier Architectures

Deploying classifiers inside modern desktop browsers via @huggingface/transformers (Transformers.js) eliminates ongoing infrastructure costs. Instead of massive autoregressive models, the pipeline employs fine-tuned sequence-classification encoders:

- **Microsoft DeBERTa-v3-small:** Leverages disentangled attention, computing attention weights across token content and relative position vectors independently. This separation makes the model sensitive to synthetic structural regularities and unnatural token transitions.
- **RoBERTa-base AI Detector:** Quantized for token- and sentence-classification to generate probability curves across sliding sequence windows.

Quantization Profiles and On-Device Execution

Executing transformer models locally in the browser requires post-training quantization to reduce download sizes and memory demands:

- **8-bit Quantization (W8A8 / INT8):** Graph-optimized checkpoints run via ONNX Runtime Web using WebAssembly with SIMD acceleration.

An INT8 DeBERTa-v3-small model occupies approximately 135 MB, retaining within 0.8% of full-precision AUROC performance.

- **4-bit Quantization (INT4 / WebGPU):** Using the WebGPU execution provider allows INT4 weight models to fit within a 72 MB footprint. This setup processes an average 1,500-word essay across 7 overlapping inference windows in 1.2 to 2.4 seconds on modern consumer hardware, requiring zero cloud compute.

Sliding Window Stride Inference Algorithm

Because students interweave generated text with original writing, evaluation operates across continuous sliding windows. Text is segmented into lexical sentences, mapped directly to absolute character offsets $[C_{\text{start}}(j), C_{\text{end}}(j)]$. Token sequences are processed using a window length $W = 256$tokens with a stride $S = 128$ tokens.

For tokens located within overlapping windows, individual window probability predictions are merged using a Gaussian distance-weighting function centered on each window:

$$P_{\text{AI}}(t_i) = \frac{\sum_m w_m\,(i) \cdot P_{\text{AI}}^{(m)}(t_i)}{\sum_m w_m\,(i)}, \quad w_m(i) = \exp\left(-\frac{(i - c_m)^2}{2\sigma_w^2}\right)$$

This weighting eliminates sharp probability boundaries at window seams, producing smooth character-level probability curves across the document.

Forensic File Artifacts and Metadata Inspection

Document container forensics offers non-probabilistic, deterministic evidence. Statistical text markers indicate generative patterns, but structural metadata can confirm copy-paste operations, external scripting, and impossible writing timelines.

DOCX OpenXML Container Architecture

A Microsoft Word .docx file is a compressed PKZip archive adhering to the Ecma-376 Office Open XML standard. Extracting the archive reveals structural XML parts that record editing histories and session data:

- word/document.xml: Contains the primary document text, split into paragraphs (<w:p>) and text runs (<w:r>), decorated with revision tracking attributes.

- word/settings.xml: Records the rsidRoot and the master rsidTable, an eight-digit hexadecimal registry of every save session associated with the file.
- docProps/app.xml and docProps/core.xml: Store document-level summary metadata, including active editing duration, creation dates, and revision counters.

Revision Save Identifiers and Paste Ingestion

Microsoft Word assigns an eight-digit hexadecimal Revision Save Identifier (RSID) to every editing session that begins and ends with a save action. Specific XML attributes trace these edits:

- w:rsidRoot: Marks the document's initial generation session.
- w:rsidR: Identifies the session that inserted a specific run of text.
- w:rsidP: Identifies the session that altered paragraph-level formatting.
- w:rsidRDefault: Defines the default editing session for newly added children within a paragraph.

Authentic student writing generates a dense, varied RSID history. As a student types, corrects typos, inserts phrases, and revises sections over multiple days, Word generates dozens to hundreds of distinct entries in the rsidTable. Paragraphs contain multiple interlaced text runs (<w:r>) carrying divergent w:rsidR values.

When a student copies synthetic output from a browser and pastes it into Word, the entire text block is assigned a single, uniform w:rsidR value. If an 8-page document displays only one or two RSIDs across all body paragraphs, the contents were inserted as a single monolithic block.

Furthermore, if the paragraph properties retain an rsid inherited from an underlying template (w:rsidP) while the text run displays a completely disconnected w:rsidR, the discrepancy confirms that external text was pasted over the template structure.

Editing Timelines and Physical Typing Limits

The docProps/app.xml file contains the <TotalTime> element, which tracks the cumulative minutes Word was open and actively focused on the document. The docProps/core.xml file tracks the revision counter (<cp:revision>) alongside UTC creation and modification timestamps.

In authentic composition, typing a 2,500-word academic essay requires multiple hours of active editing across multiple sessions. In plagiarized submissions, forensics frequently reveals a 2,500-word document with a <TotalTime> of 0 or 1 minute and a <cp:revision> count of 1.

Producing 2,500 words in 60 seconds exceeds physical human typing capabilities, providing deterministic proof of external generation and pasting.

PDF Internal Stream and Engine Forensics

When documents are submitted as PDFs, OpenXML session logs are absent, but low-level PostScript structures reveal generation mechanisms:

- **Producer and Creator Signatures:** The /Info dictionary and XMP metadata stream expose the generation pipeline. Documents compiled through automated headless scrapers or programmatic wrappers display telltale /Producer signatures—such as Skia/PDF m120 (Google Chrome headless print-to-PDF), wkhtmltopdf, WeasyPrint, or ReportLab. Standard student submissions instead display consumer office software (e.g., Microsoft® Word for Microsoft 365 or Quartz PDFContext).
- **Font Subsetting Structures:** Standard word processors subset embedded fonts using PostScript naming conventions, prefixing them with a randomized six-letter tag and a plus sign: [A-Z]{6}\+[FontName] (e.g., ABCDAA+Calibri-Bold). Programmatic PDF generators frequently omit these tags, embed whole font packages, or generate mismatched glyph width dictionaries (/Widths [...]).
- **Modification Delta Inconsistencies:** The forensic delta $\Delta t = |\text{ModDate} - \text{CreationDate}|$ exposes anomalous processing workflows. Legitimate human writing exhibits iterative saves, whereas synthetic or converted documents often present simultaneous creation and modification dates, or modification dates that predate creation timestamps due to unsynchronized script execution.
- **Incremental Update XREF Tables:** Iterative manual editing creates an appended cross-reference table (xref) for each save action. A long, complex PDF featuring a single, completely serialized xref table indicates that the document was generated and compiled in a single automated step.

Adversarial and Paraphrase Resilience

Students regularly use evasion techniques—such as synonym substitution tools (QuillBot) or adversarial character injection—to bypass detection

algorithms. Submissions must pass through an automated sanitization and forensic inspection pipeline before linguistic analysis.

Character Injection Strategies

- **Zero-Width Character Embedding:** Inserting invisible formatting characters into words (e.g., inserting zero-width spaces into computer) breaks string-matching and n-gram tokenization. Common injection characters include:
  - Zero-Width Space (U+200B)
  - Zero-Width Non-Joiner (U+200C)
  - Zero-Width Joiner (U+200D)
  - Zero-Width No-Break Space (U+FEFF)
  - Word Joiner (U+2060)
- **Cross-Script Homoglyph Substitution:** Replacing Latin characters with visually indistinguishable characters from Cyrillic, Greek, or Cherokee Unicode blocks (e.g., replacing Latin e (U+0065) with Cyrillic е (U+0435)).
- **Mathematical Alphanumeric Variations:** Using stylized mathematical letters (e.g., 𝑈 (U+1D448)) to disrupt neural subword tokenizers like Byte-Pair Encoding (BPE).

Input Sanitization and Evasion Traps

Submissions are processed through a multi-stage sanitization pipeline prior to analysis:

1. **Unicode Normalization Form KC (NFKC):** Decomposes compatibility characters and maps fullwidth, stylized, and mathematical variants back to standard canonical representations.
2. **Targeted Strip of Non-Printing Characters:** Removes all non-printing control characters using targeted regular expressions ([\u200B-\u200D\uFEFF\u2060\u00AD]).
3. **Homoglyph Mapping:** Scans tokens against a lookup table that converts lookalike scripts back to Latin-ASCII characters.
4. **Adversarial Evasion Flagging:** Rather than discarding these anomalies, the system records the frequency and locations of stripped characters. Detecting more than three zero-width spaces or foreign homoglyphs triggers an automated evasion warning, providing strong corroborating evidence for disciplinary reviews.

Paraphrase Analysis

Automated paraphrasing tools disrupt exact n-gram matching, but introduce secondary artifacts across linguistic structures:

- **Curvature Invariance:** Fast-DetectGPT's conditional probability curvature is robust against paraphrasing attacks, as rewritten text typically retains the global semantic entropy profile of the underlying model.
- **Semantic Drift Analysis:** Paraphrasing engines frequently introduce subtle context mismatches through synonym substitution. Evaluating adjacent sentences with a compact embedding model (all-MiniLM-L6-v2) identifies these breaks:

$$\text{Sim}(s_j, s_{j+1}) = \frac{\mathbf{e}_j \cdot \mathbf{e}_{j+1}}{\|\mathbf{e}_j\| \|\mathbf{e}_{j+1}\|}$$

While natural human writing maintains smooth semantic transitions, rule-based paraphrasing produces erratic drops in inter-sentence similarity.

Mitigating False Positives

Accusing a student of academic dishonesty carries serious consequences. Detection systems must maintain a near-zero false-positive rate. Standard AI detectors routinely misidentify non-native English speakers (ESL/EFL) and neurodivergent writers due to structural biases in perplexity scoring.

ESL/EFL Syntactic Traps and Biases

Non-native English writers often use a more focused vocabulary and rely heavily on formulaic transition phrases ("Furthermore," "In addition," "It is important to remember"). Because these patterns align closely with the high-probability token paths generated by language models, statistical detectors regularly assign them low perplexity, misclassifying genuine student writing as AI-generated.

Protective Verification Rules

To protect ESL/EFL and neurodivergent students, the system implements specific verification checks:

- **Linguistic Variance Thresholding:** Pure perplexity cannot trigger a synthetic text flag on its own. Low perplexity only contributes to an alert if accompanied by low burstiness and an elevated neural classifier probability.

- **Syntactic Discrepancy Gating:** Native AI output pairs low perplexity with correct grammar, uniform register, and canonical collocations. ESL student writing pairs low perplexity with non-standard collocations, awkward prepositions, and grammatical errors. Detecting these genuine grammatical inconsistencies acts as a hard veto that suppresses the synthetic probability score.
- **Minimum Span Length Thresholds:** Analysis is disabled for text segments under 150 consecutive tokens. Over short spans, human academic writing frequently exhibits low perplexity by chance (e.g., in literature citations, direct definitions, or standard lab descriptions), which can trigger false positives.
- **Neurodivergent Cadence Protection:** Highly structured, repetitive phrasing is evaluated using burstiness across the entire document rather than isolated paragraphs, ensuring that consistent individual writing styles are not misclassified.
- **Mandatory Multi-Signal Corroboration:** Statistical text markers alone cannot support formal disciplinary action. The platform requires corroboration across two independent forensic categories—such as an elevated neural classifier score combined with an OpenXML metadata anomaly (e.g., zero editing time or a single monolithic RSID cluster).

Image and Diagram Detection

Generative visual models—including Latent Diffusion Models (Stable Diffusion), DALL-E, and Midjourney—are widely used to generate artwork, diagrams, presentation slides, and figures. Verifying visual submissions requires inspecting cryptographic provenance credentials, analyzing frequency-domain rendering artifacts, and identifying semantic inconsistencies in embedded text and layout geometry.

Digital Provenance and Watermark Parsing

Cryptographic provenance and embedded watermarks provide verifiable indicators of generative origin without relying on probabilistic computer vision models.

Coalition for Content Provenance and Authenticity Framework

The C2PA standard, developed by the Content Authenticity Initiative (CAI), defines an open framework for embedding verifiable provenance data into digital media files. Leading generative platforms (such as OpenAI's DALL-E 3,

Adobe Firefly, and Microsoft Designer) embed cryptographically signed C2PA manifest stores directly into JPEG, PNG, and WebP containers.

A C2PA manifest store is embedded within a dedicated container box (in JPEG, an APP11 segment containing a JUMBF box—JPEG Universal Metadata Box Format; in PNG, a c2pa chunk). The manifest contains three core elements:

- **Assertions:** Statements of fact detailing the asset's creation. Generative platforms include a c2pa.actions.v2 assertion with an action of c2pa.created and a digitalSourceType set to https://cv.iptc.org/newscodes/digitalsourcetype/trainedAlgorithmicMedia.
- **Claim:** An aggregation of hashed assertions bound to the underlying media data using a hard-binding SHA-256 hash across the raw pixel byte stream.
- **Claim Signature:** An X.509 PKI cryptographic signature generated via the platform's private key, chained to an approved public root certificate authority.

The system uses the client-side library @contentauth/c2pa-web via a WebAssembly module (c2pa.wasm). The verification process executes entirely in the teacher's browser without transferring the image to an external server:

1. The browser reads the image file buffer as an ArrayBuffer.
2. createC2pa() scans the byte stream for the JUMBF marker block.
3. If present, the digital signature is validated against an embedded list of trusted certificate roots (Adobe, OpenAI, Microsoft, Google).
4. The action history is parsed. If trainedAlgorithmicMedia or compositeWithTrainedAlgorithmicMedia is discovered, the asset is confirmed as synthetic.

Google SynthID Invisible Watermarking

Google's SynthID embeds an imperceptible watermark into the high-frequency components of generated media (used across Imagen and Gemini). The system applies pseudo-random coordinate perturbations across the frequency spectrum during image synthesis.

While the proprietary decoding key is held by Google, open-source Bayesian and frequentist detectors can detect these frequency-domain alignments. The detector computes spatial G-values across the image and runs a binomial hypothesis test against the unwatermarked null hypothesis, returning an

exact p-value. If $p < 10^{-6}$, the submission contains an authentic SynthID watermark.

Text Chunks and EXIF Parameters

Open-source image synthesis pipelines leave metadata traces in container headers:

- **PNG Text Chunk Extraction (tEXt, zTXt, iTXt):** Web user interfaces such as Automatic1111, ComfyUI, and InvokeAI write raw generation parameters directly into image metadata. A script scans the image byte buffer for the parameters or prompt chunk keywords. Discovering fields like Steps:, Sampler:, CFG scale:, or Model hash: provides definitive proof that the image was synthesized.
- **EXIF Tag Traces:** Generative platforms frequently leave traces in standard EXIF tags (such as ImageDescription, Software, or UserComment). Midjourney historically embedded generation parameters in EXIF tags, while Adobe Firefly flags specific XMP credit metadata.

Frequency-Domain and Artifact Forensics

When an image lacks C2PA credentials or has had its metadata stripped by re-saving or screenshotting, the system evaluates spatial and frequency-domain artifacts left by generative rendering pipelines.

Azimuthally Averaged Power Spectrum and Upsampling Grids

Natural camera captures and hand-drawn illustrations adhere to consistent statistical properties. In particular, the radially averaged power spectrum of natural images follows a strict power-law falloff governed by the scale-invariance principle:

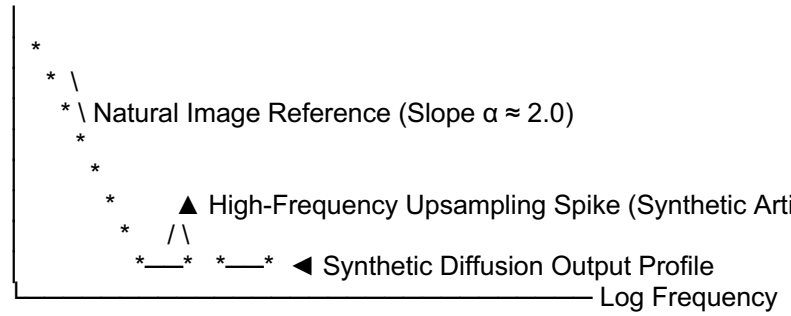
$$P(f_r) \propto f_r^{-\alpha}, \quad \text{where } \alpha \approx 2.0$$

Here, $f_r = \sqrt{u^2 + v^2}$ represents the radial spatial frequency.

Diffusion models and generative adversarial networks (GANs) break this power law due to their underlying neural architectures. The generation pipeline uses upsampling layers (such as transposed convolutions or nearest-neighbor scaling), which introduce periodic grid artifacts into the output.

Radially Averaged Power Spectrum: Log Power vs. Log Frequency

Log Power



Client-Side 2D FFT Analysis Pipeline

Using a lightweight WebAssembly FFT implementation (such as KissFFT compiled to WASM), the system evaluates images locally:

1. Standardize the input image to a 512×512 grayscale patch $I(x, y)$.
2. Apply a 2D Hann window to suppress edge discontinuities.
3. Compute the 2D Fast Fourier Transform:

$$\mathcal{F}(u, v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} I\,(x, y) w(x, y) e^{-i\frac{2\pi}{N}(ux+vy)}$$

4. Calculate the power spectrum $S(u, v) = |\mathcal{F}(u, v)|^2$.
5. Perform azimuthal averaging to calculate the 1D radial profile $P(f_r)$ across concentric rings.
6. Fit a linear regression to the log-log transform $\log P\,(f_r) = -\alpha \log f_r + c$ over the mid-to-high frequency bands.
7. Calculate the variance of high-frequency residuals. Diffusion outputs introduce distinct energy spikes at harmonic grid intervals, causing the fitted slope α to deviate significantly from the natural range ($1.8 \leq \alpha \leq 2.2$).

Chromatic Aberration and Color Space Distributions

Physical camera lenses suffer from optical dispersion, causing wavelength-dependent refraction that leaves subtle, uniform chromatic aberration along high-contrast radial edges. Diffusion models synthesize red, green, and blue color channels simultaneously within a learned latent space (such as Stable Diffusion's 4-channel VAE). As a result:

- Synthetic images display zero physical chromatic aberration along high-contrast edges.
- Cross-channel covariance displays localized correlation artifacts across flat regions.

- Residuals from high-pass filtering show that generative images lack the Poisson-distributed photon shot noise characteristic of physical camera sensors, displaying instead smooth, spatially uniform noise profiles punctuated by deconvolution lattice patterns.

Text-in-Image and Semantic Inconsistencies

Generative images of diagrams, charts, and infographics frequently struggle to render coherent text and maintain structural geometry.

Client-Side OCR and Gibberish Evaluation

Diagrams submitted for technical or scientific assignments can be evaluated by extracting embedded text using Tesseract.js (an open-source Tesseract OCR engine compiled to WebAssembly).

1. **Text Box Extraction:** The WASM OCR engine detects bounding boxes and transcribes text elements across the image.
2. **Lexical Coherence Check:** Extracted strings are evaluated against a standard English dictionary alongside relevant domain lexicons. While older diffusion models produce illegible scribbles, newer models often output believable pseudo-words (e.g., "Phorosynthesis" or "Mitocondrial").
3. **Character N-Gram Entropy:** The character sequence is evaluated using a character-level tri-gram model trained on academic text. A high gibberish score ($P_{\text{gibberish}} > 0.40$) indicates synthetic text generation, as human illustrators rarely introduce garbled, pseudo-lexical characters into diagrams.

Geometric Inconsistencies and Vector Logic

Diagrams generated by AI models often lack structural and geometric consistency:

- **Axis and Grid Line Orthogonality:** The system applies a Hough Transform over edge-detected diagrams (using Canny filtering) to extract linear vectors. In authentic scientific plots, gridlines and axes intersect at precise 90-degree angles. AI-generated charts frequently exhibit floating grid lines, non-parallel ticks, and vanishing axes.
- **Flowchart Directed Logic:** Using basic contour tracing, the system validates node-link topologies. Generative images regularly produce arrows that terminate in empty space, lines that pass through text boxes

without connection nodes, and flowchart shapes that blend into one another.

UI Data Modeling and Aggregation Strategy

To support high-stakes disciplinary decisions, the application cannot rely on single, opaque detection scores. Instead, it must synthesize multi-modal signals into a deterministic, explainable evaluation pipeline.

Deterministic Multi-Signal Confidence Fusion Engine

The Confidence Fusion Engine combines statistical text metrics, neural classifier spans, and forensic container metadata into a unified score. The engine uses a log-odds scoring model with structural veto gates to keep the false-positive rate near zero.

Log-Odds Scoring Model

Let each independent forensic module produce an evidence score $s_k \in [0,1]$. The scores are transformed into log-odds space:

$$\ell_k = \log\left(\frac{s_k + \epsilon}{1 - s_k + \epsilon}\right)$$

The aggregate log-odds L_{total} is computed as a weighted combination incorporating a prior base rate ℓ_0:

$$L_{\text{total}} = \ell_0 + \sum_k w_k \cdot \ell_k$$

The resulting aggregate score maps back to a calibrated probability $P_{\text{aggregate}}$:

$$P_{\text{aggregate}} = \frac{1}{1 + \exp(-L_{\text{total}})}$$

The evaluator applies calibrated weights reflecting the evidentiary reliability of each signal:

- $w_{\text{c2pa}} = 10.0$: A verified C2PA generative claim functions as definitive cryptographic proof.
- $w_{\text{meta}} = 3.5$: Structural metadata anomalies (e.g., ≤ 1 minute editing time on a 2,000-word file, or a single monolithic RSID cluster) provide strong forensic evidence.

- $w_{\text{neural}} = 2.5$: Transformer-based span classification (DeBERTa-v3/RoBERTa) captures localized generative phrasing.
- $w_{\text{bino}} = 2.0$: Binoculars normalized cross-perplexity ratio measures statistical distribution shifts.
- $w_{\text{burst}} = 1.5$: Negative sentence-level burstiness identifies flat cadence.
- $w_{\text{gltr}} = 1.0$: GLTR rank bin distribution ($T_{\text{top10}} > 0.85$) provides supplementary token-level evidence.

Hard-Veto and Override Rules

To prevent false accusations, the system applies hard-coded structural overrides:

- **The Minimum Length Veto:** Any continuous span of fewer than 150 words is clamped to Tier 0 (Insufficient Evidence), regardless of neural classifier outputs.
- **The ESL Protection Veto:** If the text exhibits a low Type-Token Ratio alongside frequent grammatical, prepositional, or tense errors, the system suppresses pure statistical signals ($w_{\text{bino}}, w_{\text{burst}}$), setting their weights to 0.0. The final score is then capped at Tier 1 unless corroborating metadata anomalies (such as a 0-minute editing duration) are present.
- **The Metadata-Absent Downgrade:** If an uploaded file lacks internal session tracking (e.g., an unformatted .txt file or a flattened PDF), the system caps confidence at Tier 2 (Highly Probable). Disciplinary actions categorized under Tier 3 require explicit forensic confirmation (e.g., cryptographic provenance or an OpenXML editing trail).

Tiered Confidence Classifications

- **Tier 0 (Green — Human-Consistent / Low Risk):** $P_{\text{aggregate}} < 0.35$. Typical human writing dynamics, normal burstiness profiles, or insufficient text length for definitive evaluation.
- **Tier 1 (Yellow — Inconclusive / AI-Assisted):** $0.35 \leq P_{\text{aggregate}} < 0.70$. Minor statistical anomalies, formulaic text spans, or light grammar-tool assistance. Suitable for educational check-ins, but excluded from formal academic misconduct proceedings.
- **Tier 2 (Orange — Highly Probable AI):** $0.70 \leq P_{\text{aggregate}} < 0.95$. Low perplexity, negative burstiness, and high-probability neural classifier spans. Flags the submission for human instructor review.

- **Tier 3 (Red — Proven Synthetic):** $P_{\text{aggregate}} \geq 0.95$. Supported by deterministic evidence—such as a valid C2PA generative claim, a 0-minute total editing time on a long document, or monolithic RSID insertions across the entire paper.

Production-Ready JSON Data Contract

The engine returns a standardized JSON payload that maps directly to UI text highlights and progressive-disclosure evidence drawers.

TypeScript

```
/**
 * Root response object returned by the multi-modal detection engine.
 */
interface AnalysisReportPayload {
  submission_id: string;
  timestamp: string; // ISO 8601
  overall_verdict: {
    tier: 0 | 1 | 2 | 3;
    tier_label: "Human-Consistent" | "Inconclusive" | "Likely Synthetic" | "Proven Synthetic";
    aggregate_score: number; // [0.00, 1.00]
    disciplinary_action_eligible: boolean; // True ONLY if Tier 3 or (Tier 2 + corroborating flags)
    summary_justification: string;
  };
  document_metadata_forensics: {
    file_type: "docx" | "pdf" | "image" | "plain";
    editing_duration_minutes: number | null;
    revision_count: number | null;
    author_tool_signature: string;
    anomaly_detected: boolean;
    flags: Array<{
      flag_code: string;
      description: string;
      severity: "low" | "medium" | "critical";
    }>;
  };
  image_forensics?: {
    c2pa_manifest_present: boolean;
    c2pa_verified_ai_generator: string | null;
    exif_software_tag: string | null;
    frequency_spectral_slope_alpha: number | null;
    ocr_gibberish_detected: boolean;
  };
  highlighted_spans: Array<TextHighlightSpan>;
}

/**
 * Span-level attribution segment for UI highlighting and progressive disclosure.
 */
interface TextHighlightSpan {
  id: string;
  start_char: number; // 0-based character offset
```

```
  end_char: number;   // 0-based character offset
  text_segment: string;
  severity_tier: 1 | 2 | 3;
  confidence: number; // [0.00, 1.00]
  reason_code: "MONOLITHIC_INSERTION" | "LOW_BURSTINESS_UNIFORM_PPL" |
"NEURAL_SPAN_POSITIVE" | "HOMOGLYPH_EVASION";
  forensic_metrics: {
    sentence_burstiness: number;
    binoculars_score: number;
    gltr_top10_percentage: number;
    associated_rsid?: string;
    stripped_adversarial_chars?: number;
  };
  esl_protective_override_applied: boolean;
}
```

The corresponding concrete JSON instance returned by the endpoint structure populates the frontend interface:

JSON


```
{
  "submission_id": "sub_academic_2026_09842",
  "timestamp": "2026-11-03T16:42:00.000Z",
  "overall_verdict": {
    "tier": 3,
    "tier_label": "Proven Synthetic",
    "aggregate_score": 0.982,
    "disciplinary_action_eligible": true,
    "summary_justification": "Submission demonstrates monolithic text insertion with zero editing time,
corroborated by high-confidence neural span classification and negative burstiness."
  },
  "document_metadata_forensics": {
    "file_type": "docx",
    "editing_duration_minutes": 1,
    "revision_count": 1,
    "author_tool_signature": "Microsoft Office Word (Windows)",
    "anomaly_detected": true,
    "flags": [
      {
        "flag_code": "DOCX_IMPOSSIBLE_WPM",
        "description": "Document contains 2,412 words across 6 pages but records only 1 minute of active
editing time.",
        "severity": "critical"
      },
      {
        "flag_code": "SINGLE_RSID_TREE",
        "description": "Over 95% of the body text was inserted under a single Revision Save Identifier
(rsidR='00DA49F2').",
        "severity": "critical"
      }
    ]
  },
  "highlighted_spans": [
    {
      "id": "span_01",
      "start_char": 420,
```

```
      "end_char": 1890,
      "text_segment": "The advancement of CRISPR-Cas9 genome editing mechanisms has altered
contemporary oncology...",
      "severity_tier": 3,
      "confidence": 0.991,
      "reason_code": "MONOLITHIC_INSERTION",
      "forensic_metrics": {
        "sentence_burstiness": -0.42,
        "binoculars_score": 0.741,
        "gltr_top10_percentage": 0.884,
        "associated_rsid": "00DA49F2",
        "stripped_adversarial_chars": 0
      },
      "esl_protective_override_applied": false
    }
  ]
}
```


Edge and Client-Side Deployment Architecture

To operate under zero-cost constraints, the architecture uses a hybrid pipeline that pairs modern browser capabilities with serverless edge workers. Processing tasks are distributed between the teacher's local machine and a free-tier edge network.

Document parsing and container analysis are executed directly in the browser using JSZip for DOCX archives and pdfjs-dist for PDF stream inspection. Extracting XML nodes, calculating RSID distributions, and verifying editing durations requires under 50 ms of CPU time and incurs no server infrastructure costs.

Neural classification is performed locally using @huggingface/transformers running via ONNX Runtime Web. The quantized INT8 model checkpoint (~75 MB) is downloaded once and stored in the browser's persistent CacheStorage or IndexedDB. When a teacher opens an assignment, inference runs locally using the client machine's WebGPU or multi-threaded WASM SIMD, keeping cloud compute costs at zero.

Cloudflare Workers (operating within free-tier limits of 128 MB RAM and 10 ms CPU time) handle text sanitization, Unicode NFKC cleaning, zero-width character detection, and lightweight regex-based stylometric heuristics. C2PA verification is performed on-device via @contentauth/c2pa-web and c2pa.wasm, validating JUMBF containers and certificate chains locally in milliseconds. Image spectral profiling is processed in-browser via a WebGL fragment shader or KissFFT compiled to WebAssembly, calculating radial power spectrum curves and fitting the spectral slope α in approximately 15 ms.

Pipeline Stage	Execution Runtime	Dependencies	Memory Footprint	Latency	Infrastructure Cost
DOCX/PDF Container Parsing	Client Browser (JS/DOM)	JSZip, pdfjs-dist	≈ 25 MB	$30 - 80$ ms	$0.00
Unicode Sanitization	Edge Worker / Client	Vanilla ES6 RegExp	< 2 MB	< 5 ms	$0.00 (Free Tier)
Neural Span Classifier	Client (WebGPU / WASM)	Transformers.js (ONNX)	≈ 180 ME	$800 - 2400$ ms	$0.00
Statistical & Burstiness	Edge Worker / Client	Native Math Functions	< 10 MB	$15 - 40$ ms	$0.00
C2PA Manifest Parsing	Client Browser (WASM)	@contentauth/c2pa-web	≈ 35 MB	$40 - 120$ ms	$0.00
2D FFT Image Forensics	Client Browser (WebGL/WASM)	KissFFT / Custom WebGL	≈ 30 MB	$15 - 45$ ms	$0.00
Tesseract.js OCR (Diagrams)	Client Web Worker	tesseract.js-core(WASM)	≈ 120 ME	$1200 - 3500$ ms	$0.00

Defensive Academic Integrity Protocols

To support academic integrity hearings and parental reviews, the system avoids black-box probability scores in favor of clear, verifiable chains of evidence.

Evidence Presentation Chain:

1. Non-Probabilistic Container Metadata
   - Verify active editing duration (TotalTime) against physical typing limits.
   - Inspect RSID session logs to detect monolithic external paste insertions.

2. Corroborating Linguistic Analysis
   - Evaluate sentence burstiness dispersion (B < -0.10) to confirm artificial cadence.
   - Inspect GLTR rank partitions (Top-10 > 80%) and token-level neural classification.

3. Anti-Bias and Demographic Protection Checks
   - Verify the absence of genuine ESL grammatical, tense, or collocation errors.
   - Confirm segment length exceeds minimum thresholds (N >= 150 tokens).

4. Cryptographic Provenance and Direct Artifacts
   - Validate C2PA PKI certificate signatures for generative source declarations.
   - Extract raw PNG generation parameters and evaluate frequency-domain grid anomalies.

Protocol for Academic Misconduct Hearings

When an analysis flags a submission for potential disciplinary action, the application generates a comprehensive forensic summary designed to withstand scrutiny:

- **Lead with Non-Probabilistic Container Evidence:** Rather than relying on a statistical probability score, reports begin with verifiable file metadata. For example: "The submitted DOCX file contains 2,800 words across 8 pages, but records only 1 cumulative minute of active editing time. Furthermore, 100% of the body text was inserted under a single Revision Save Identifier (rsidR='00E41A12'), confirming that the text was written externally and pasted into the document in a single action".
- **Layer in Corroborating Linguistic Metrics:** Reports supplement structural evidence with statistical markers: "Sentence-level burstiness is strongly negative ($B = -0.48$), indicating an artificially uniform syntactic cadence. Token rank analysis shows that 89% of all words fall within the top-10 most probable model predictions, a distribution characteristic of generative language models and rare in human academic writing".
- **Document Demographic Protection Checks:** Reports verify that ESL protections were applied: "The text was evaluated for second-language acquisition patterns. The absence of characteristic grammatical, prepositional, and spelling errors confirms that low lexical entropy is not the result of an ESL writing profile."
- **Present Cryptographic Provenance:** For visual submissions containing verified C2PA manifests, reports provide the complete digital certificate validation trail: "Image asset asserts trainedAlgorithmicMedia via C2PA manifest signed by OpenAI Public Certificate Authority, confirming synthetic image generation".

By synthesizing structural container forensics, client-side neural classification, and strict bias-protection rules, the architecture delivers an in-house detection platform that remains defensible in disciplinary reviews, runs entirely on free and on-device resources, and prevents false accusations against non-native English writers.